Soft constraint | NRMSE summary (util.py: RMS normalization)

Median of per-IC NRMSE on the same 7 complete test ICs for every model

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	baseline	0.15	0.476	1.02	1.02	0.329	5.49	0.509	0.967	0.997	0	18/18
1 ANN10	soft10k	0.323	0.9	10.6	8.44	0.613	7.79	0.585	163	2.13e+03	0	14/18
1 ANN10	softlong	0.813	0.979	1	73.9	0.628	1	1	350	602	0	16/18
2 P-only DeepONet10	baseline	0.0551	0.236	1	1	0.0818	1	0.999	0.997	1	0	18/18
2 P-only DeepONet10	soft10k	0.52	0.854	1	1	0.909	1	1	504	970	0	18/18
2 P-only DeepONet10	softlong	0.313	0.971	11	4.76	0.623	3.25	0.889	79.3	1.32e+03	0	16/18
3 ANN10 + P-only DeepONet10	baseline	5.60e-03	0.233	1	1	0.0322	0.998	0.999	0.992	0.999	0	18/18
3 ANN10 + P-only DeepONet10	soft10k	0.301	0.867	10.3	10.9	0.538	7.59	0.601	700	1.22e+04	0	18/18
3 ANN10 + P-only DeepONet10	softlong	0.545	0.969	19.9	29.4	0.691	1	0.982	541	2.95e+03	0	18/18
4 T/P DeepONet10	baseline	0.0406	0.212	2.92	0.887	0.196	0.952	0.626	0.993	0.997	0	17/18
4 T/P DeepONet10	soft10k	0.6	0.909	9.45	3.92	0.628	3.4	0.999	59.3	1	0	15/18
4 T/P DeepONet10	softlong	0.474	0.985	2.47	26.4	0.525	3.85	0.983	110	7.73e+03	0	14/18
5 ANN10 + T/P DeepONet10	baseline	0.0362	0.13	2.09	0.965	0.13	0.999	0.625	0.921	0.99	0	18/18
5 ANN10 + T/P DeepONet10	soft10k	0.342	0.968	11	9.17	0.631	2.07	0.656	139	372	0	18/18
5 ANN10 + T/P DeepONet10	softlong	0.252	0.964	3.38	13.1	0.644	1.41	0.667	54.7	137	0	18/18
6 T/P/ϕ DeepONet10	baseline	0.542	0.931	1	1	0.733	1	1	1	1	0	17/18
6 T/P/ϕ DeepONet10	soft10k	0.474	0.976	9.05	4.13	0.598	1.44	1	14.7	3.79	0	17/18
6 T/P/ϕ DeepONet10	softlong	0.463	0.99	11.9	23.2	0.527	4.51	0.989	402	3.46e+03	0	16/18
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.542	0.888	1	1	0.733	1	1	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.354	0.952	6.19	5.75	0.621	2.14	0.651	161	375	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.323	0.819	4.33	30.2	0.615	6.8	0.568	3.67	9.85	0	17/18

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
Common ICs: 5, 7, 39, 44, 53, 64, 69.